

# First Lesson Walkthrough

## Goal

Run one complete learning loop: normal story training → same prompt comparison → pirate-style retrain → evidence-based reflection.

By the end, students should be able to say:

- an LLM predicts the next token from earlier tokens
- training data changes output patterns
- lower loss means lower prediction error, not human understanding
- attention is useful to inspect, but it is not proof of thinking

## Sample datasets for this lesson

- Normal style dataset: `data/samples/space_adventure.txt`
- Retrain style dataset: `data/samples/pirate_dialogue.txt`

Keep the prompt the same in both generation steps. That makes the dataset change easier to see.

## What success looks like

- The model trains without errors.
- The first output resembles normal story text.
- Pirate retrain output shows some pirate vocabulary or rhythm.
- Learners can explain that changed data changed model behavior.

## Before/after comparison table

| Stage                | Prompt                            | Expected style              | What learners should notice                                             |
|----------------------|-----------------------------------|-----------------------------|-------------------------------------------------------------------------|
| Before retrain       | Captain Rowan looked at the stars | Neutral or space-story tone | Word choices are closer to the normal story dataset.                    |
| After pirate retrain | Captain Rowan looked at the stars | Pirate-flavored tone        | Terms like “matey,” “sail,” or “tide” may appear, and rhythm may shift. |

## Step 1 — Train on normal story data

```
kairo-train --input_file data/samples/space_adventure.txt --out_dir
runs/lesson_normal --epochs 1 --batch_size 4 --seq_len 32 --d_model 64 --n_heads 4
--n_layers 2 --device cpu
```

Ask students to notice:

- the number of parameters
- the train and validation loss
- the best .pt checkpoint path

## Step 2 — Generate output before retrain

```
kairo-generate --checkpoint runs/lesson_normal/best.pt --prompt "Captain Rowan  
looked at the stars" --max_new_tokens 30 --device cpu
```

### Example output (before retrain)

```
Captain Rowan looked at the stars and the station lights flickered in the dark  
hall
```

Record the real classroom output even if it is odd. Tiny models are allowed to be imperfect; that imperfection makes the mechanism easier to discuss.

## Step 3 — Retrain on pirate dataset

```
kairo-train --input_file data/samples/pirate_dialogue.txt --out_dir  
runs/lesson_pirate --epochs 1 --batch_size 4 --seq_len 32 --d_model 64 --n_heads 4  
--n_layers 2 --device cpu
```

In the command-line lesson, "retrain" means training the same tiny architecture again with different data and then comparing behavior. Learn Mode keeps the before/after comparison in the same interactive session.

## Step 4 — Generate output after retrain

```
kairo-generate --checkpoint runs/lesson_pirate/best.pt --prompt "Captain Rowan  
looked at the stars" --max_new_tokens 30 --device cpu
```

### Example output (after retrain)

```
Captain Rowan looked at the stars, matey, and swore by the tide to raise the black  
sail
```

## Step 5 — Compare with evidence

| Question                        | Evidence to collect                          |
|---------------------------------|----------------------------------------------|
| Which vocabulary changed?       | New words, repeated words, style markers     |
| Did the tone change?            | Adventure, pirate, poetic, helpful, factual  |
| Did the structure change?       | Sentence length, punctuation, repetition     |
| Did the model become "smarter"? | Separate fluency from truth or understanding |

Teacher prompt:

what changed because the model saw different text, and what stayed the same because the model architecture stayed the same?

## Step 6 — Optional Learn Mode inspection

kairo-learn

Use Learn Mode to inspect:

- the byte tokens in the training text
- top next-token probabilities for the same prompt
- attention patterns for selected tokens
- before/after outputs after changing training text

## Before/after retrain exercise prompts

Captain Rowan:  
The station woke as  
Raise the black sail  
Rain taps softly  
The robot opened the door

## Exit ticket

Ask each learner to complete one sentence:

Changing the training data changed the model because...

Then ask:

One thing attention can show is...  
One thing attention cannot prove is...